

TIØ4145 Corporate Finance

Exercise 9: Financial Options

Deadline: Monday November 3rd, 2025, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday October 30th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on October 27th.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapters 20 and 21.

Tasks

Chapter 20 – Financial Options

1. You own a call option on Intuit stock with a strike price of \$40. The option will expire in exactly three months' time.
 - a. If the stock is trading at \$55 in three months, what will be the payoff of the call?
 - b. If the stock is trading at \$35 in three months, what will be the payoff of the call?
 - c. Draw a payoff diagram showing the value of the call at expiration as a function of the stock price at expiration.
2. Assume that you have shorted the call option in task 1.
 - a. If the stock is trading at \$55 in three months, what will you owe?
 - b. If the stock is trading at \$35 in three months, what will you owe?

- c. Draw a payoff diagram showing the amount you owe at expiration as a function of the stock price at expiration.

3. [Voluntary]

You own a put option on Ford stock with a strike price of \$10. The option will expire in exactly six months' time.

- a. If the stock is trading at \$8 in six months, what will be the payoff of the put?
- b. If the stock is trading at \$23 in six months, what will be the payoff of the put?
- c. Draw a payoff diagram showing the value of the put at expiration as a function of the stock price at expiration.

4. [Voluntary]

Assume that you have shorted the put option in task 3.

- d. If the stock is trading at \$8 in six months, what will you owe?
- e. If the stock is trading at \$23 in six months, what will you owe?
- f. Draw a payoff diagram showing the amount you owe at expiration as a function of the stock price at expiration.

- 5. You happen to be checking the newspaper and notice an arbitrage opportunity. The current stock price of Intrawest is \$20 per share and the one-year risk-free interest rate is 8%. A one year put on Intrawest with a strike price of \$18 sells for \$3.33, while the identical call sells for \$7. Explain what you must do to exploit this arbitrage opportunity.

- 6. You are long two calls on the same share of stock with the same exercise date. The exercise price of the first call is \$40 and the exercise price of the second call is \$60. In addition, you are short two otherwise identical calls, both with an exercise price of \$50. Plot the value of this combination as a function of the stock price on the exercise date. What is the name of this combination of options?

- 7. Wesley Corp. stock is trading for \$27/share. Wesley has 20 million shares outstanding and a market debt-equity ratio of 0.49. Wesley's debt is zero-coupon debt with a 5-year maturity and a yield to maturity of 11%.

- a. Describe Wesley's equity as a call option. What is the maturity of the call option? What is the market value of the asset underlying this call option? What is the strike price of this call option?
- b. Describe Wesley's debt using a call option.
- c. Describe Wesley's debt using a put option.

Chapter 21 – Option Valuation

1. The current price of Natasha Corporation stock is \$6. In each of the next two years, this stock price can either go up by \$2.50 or go down by \$2. The stock pays no dividends. The one-year risk-free interest rate is 3% and will remain constant. Using the Binomial Model, calculate the price of a two-year call option on Natasha stock with a strike price of \$7.
2. Eagletron's current stock price is \$10. Suppose that over the current year, the stock price will either increase by 100% or decrease by 50%. Also, the risk-free rate is 25% (EAR).
 - a. What is the value today of a one-year at-the-money European put option on Eagletron stock?
 - b. What is the value today of a one-year European put option on Eagletron stock with a strike price of \$20?
 - c. Suppose the put options in parts a and b could either be exercised immediately, or in one year. What would their values be in this case?
3. Rebecca is interested in purchasing a European call on a hot new stock, Up, Inc. The call has a strike price of \$100 and expires in 90 days. The current price of Up stock is \$120, and the stock has a standard deviation of 40% per year. The risk-free interest rate is 6.18% per year.
 - a. Using the Black-Scholes formula, compute the price of the call.
 - b. Use put-call parity to compute the price of the put with the same strike and expiration date.

Exam Question – 2018

- a. Explain the concept of Put-Call Parity using both the formula and by plotting the payoffs as a function of the stock price.
- b. Option traders often refer to “straddles” and “butterflies”. Straddle: buy call with exercise price of \$100 and simultaneously by put with exercise price of \$100.
Butterfly: Simultaneously buy one call with exercise price of \$100, sell two calls with exercise price of \$110, and buy one call with exercise price of \$120.
 - i. Draw position diagrams for the straddle and the butterfly, showing the payoffs from the investor’s net position.
 - ii. Each strategy is a bet on variability. Explain Briefly the nature of this bet.
- c. A stock’s current price is \$160, and there are two possible prices that may occur next period: \$150 or \$175. The interest rate on risk-free investments is 6% per period. Assume that a (European) call option exists on this stock having an exercise price of \$155.
 - i. How could you form a portfolio based on the stock and the call so as to achieve a risk-free hedge?
 - ii. Compute the price of the call.
 - iii. What would change if the exercise price was \$180?

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.